



ELECT MAINT Department  
SIMHADRI

**Note No :1**



**Ref:**SMTTP/ELECT MAINT/2023-24/LMI/1030356

**Date:**

25/03/2024(17:52:31)

**Subject:**Subject:REVISION OF LMI/OIN/OPS/ELEC/007 RECOMMENDED PROCEDURES TO ASSESS THE QUALITY OF INSULATION ON GENERATOR AND MOTOR STATORS.

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LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes envisaged by COS. Only reviewing done after two years.

May please approve

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**On:**25/03/2024(17:52:31) **Ref:**SMTTP/ELECT MAINT/2023-24/LMI/1030356

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LMI is reviewed w.r.t OIN. No changes envisaged.

Put up for approval please

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**On:**19/04/2024(15:36:01) **Ref:**SMTTP/ELECT MAINT/2023-24/LMI/1030356

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**On:**19/04/2024(19:32:56) **Ref:**SMTTP/ELECT MAINT/2023-24/LMI/1030356

**NATIONAL THERMAL POWER CORPORATION**

**OPERATION DIVISION FORMAL DOCUMENTATION SYSTEM**

**OPERATION INFORMATION NOTE – OIN/OPS/ELEC/007**

**Issue No. : 1    September, 1994**

**RECOMMENDED PROCEDURES TO ASSESS THE  
QUALITY OF INSULATION ON GENERATOR AND  
MOTOR STATORS**

Approved.....R.V.Shahi.....  
Director (Operation)

Date : .....03.10.94.....

Enquiries to : Executive Director, Operation Services

# **RECOMMENDED PROCEDURES TO ASSESS THE QUALITY OF INSULATION ON GENERATOR AND MOTOR STATORS**

## **SUMMARY**

This document recommends that non-destructive electrical test procedures are to be used when assessing the condition of insulation used as the main insulation on stator winding of generators and large motors, including the larger water cooled machines.

The recommended procedures will ensure reliable monitoring of the trends of the insulation quality throughout the insulation life and will allow comparison of the test results of all generators and motors of strategic importance.

These measurements are considered to be the most suitable, at the present time, to permit assessment of the degree of degradation of the main insulation systems. They are established techniques which may be used by competent engineers who have been trained in the practice and used of the measuring equipment.

# **RECOMMENDED PROCEDURES TO ASSESS THE QUALITY OF INSULATION ON GENERATOR AND MOTOR STATORS**

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## **RECOMMENDED PROCEDURES TO ASSESS THE QUALITY OF INSULATION ON GENERATOR AND MOTOR STATORS**

### **1.0 INTRODUCTION**

The tests included in the Information Note are those recommended for monitoring the operational integrity of the electrical insulation of generators and large motors.

It is thought that there is a correlation between ageing and life, but from current knowledge, the measurements made at the stator terminals of insulation quality is inadequate to predict life expectancy. Where a winding or phase indicates an advanced degree of ageing compared with a winding of equivalent life and operating experience, advice should be sought from manufacturer's. Where it is considered necessary, more specific diagnostic tests may be applied in addition to the standard tests given in this note.

### **2.0 SUPERSEDED DOCUMENTS - Nil**

### **2.0 SCOPE**

This document recommends non-destructive electrical test procedures to be used when assessing the condition of insulation used as the main insulation on stator windings of generators and large motors, including the larger water-cooled machines.

### **3.0 GUIDANCE**

#### **4.1 Section 1 – Generators**

##### **4.1.1 Recommended Measurements**

##### **4.1.1.1 Gas Cooled Generators**

- (i) Insulation Resistance (IR) and Polarisation Index (PI) tests should be used to establish the suitability of the winding to accept the application of a high voltage a.c non-destructive test.

- ii) The condition of the bulk insulation is assessed from the variation of the tangent of the loss angle ( $\tan \delta$ ) and of capacitance (C) with voltage. Tests in steps of  $0.2 V_L$  up to an agreed maximum not exceeding  $V_L$  are required.

#### **4.1.1.2 Water Cooled Generators**

- i) Insulation Resistance (IR) and, where possible, polarisation Index (PI) tests are required to establish the suitability of the winding for the application of a high voltage a.c. non-destructive test.
- ii) The condition of the bulk insulation is assessed from the variation of the tangent of the loss angle ( $\tan \delta$ ) and capacitance (C) as detailed in 4.1.1.1 (ii)  $\tan \delta$  measurements on installed large Generators, using a schering bridge or other methods, are sometimes complicated by the presence of circulating cooled water in the stator winding. If  $\tan \delta$  is measured after draining of cooling water, the results can be erroneous if the water is not drained completely. A procedure for the measurement of  $\tan \delta$  with water flowing is enclosed at Appendix-I.

#### **4.1.2 PREPARATION OF TESTS**

- 4.1.2.1 All testing is to be done by Electrical Testing Group, taking necessary safety precautions prior to conducting the test.
- 4.1.2.2 Tests should be carried out at the beginning of an outage as soon as the machine is on barring has been completed and when adequate internal temperature stability has been achieved. At this stage, the insulation is least affected by the environment and the maximum time is available for remedial action to deal with any defects which may be identified.
- 4.1.2.3 The stator terminals should be disconnected. The test will then be made on each phase in turn with the other two phases earthed.
- 4.1.2.4 Although some water-cooled generators may be drained prior to testing, testing with primary water in circuit is also possible (Appendix-I). Testing will therefore be generally with the cooling water flowing and at a target conductivity value of 1 micro-Siemen.

N.B.

If the water has been turned off at shut down, it must be allowed to flow until the target conductivity value has completely stabilised prior to testing.

#### **4.1.3 Details of tests**

##### **4.1.3.1 Insulation Resistance testing**

###### **(i) Background Information**

Measurement of the Insulation Resistance (IR) and Polarisation Index (PI) should be made where a machine has been out of service for some time, and particularly if it is situated in a damp location, to indicate whether energization or testing with an a.c. voltage is advisable.

###### **(ii) Non-water-cooled Generators**

The d.c. source is normally generated from a mains or battery powered megohmmeter of sufficient capacity, and the values recorded. The recommended level of voltage is 1kV d.c for all windings up to and including 6.6 kV, and 5kV d.c. for windings of 11 kV or greater.

###### **(iii) Generators with water cooling.**

Achieving the desired PI should be the criterion for declaring the health of the insulation. Adequate precautions needs be taken so that surface moisture on the windings is minimized during Capital Overhauls / Forced Shut Down of the Unit.

###### **(iv) Interpretation**

If values much lower than those previously encountered are obtained after correction to 40°C, serious moisture or contamination is indicated. In practice, there is normally a doubling of the IR for every 10°C, decrease in temperature where epoxy mica insulation is used. If low values are encountered, the bushings in the generator pit may be

responsible. These should be electrically separated by guarding, and the measurements repeated. If improved results are obtained, the suspect bushings must be examined-before re-energization and cleaning and/or an alternator pit dry-out should be arranged as necessary. If no improvement is achieved, a dry-out of the winding must be considered.

- (v) Generators will normally be tested at their operational gas coolant conditions, in which case a test voltage maximum of  $V_L$  is recommended. Where stator winding history indicates suspect insulation, then a test voltage value of between  $0.6 V_L$  and  $V_L$  will be agreed with Station Management.

## **4.2 Section 2 – Motors**

### **4.2.1 Recommended Measurements**

- 4.2.1.1 Insulation Resistance (IR) and Polarisation Index (PI) test should be used to establish the suitability of the winding to accept the application of the subsequent high voltage a.c. non-destructive test.
- 4.2.1.2 The condition of the bulk insulation is assessed from the variation of the tangent of the loss angle ( $\tan \delta$ ) and of capacitance (C) with voltage. Tests in steps of  $0.2 V_L$  up to an agreed maximum not exceeding  $V_L$  are required.

### **4.2.2 Preparation for Tests**

- 4.2.2.1 All testing is to be done by Electrical Testing Group taking all necessary safety precautions.
- 4.2.2.2 Tests should preferably be carried out at the beginning of an outage when the insulation is least affected by the environment and the maximum time is available for remedial action to deal with any defects which may be identified.
- 4.2.2.3 Wherever possible, the three phases should be split at the neutral. If this cannot be achieved, a test with the three phases in parallel is then used. It is also preferred that the cables from the switchgear should be disconnected, but testing with the cables may be possible in some cases.

### **4.2.3 Details of Tests**

#### **4.2.3.1 Insulation Resistance Testing**

(i) Background Information

Measurement of the Insulation Resistance (IR) and Polarisation Index (PI) should be made where a machine has been out of service for some time, and particularly if it is situated in a damp location, to indicate whether either energization or testing with a.c. voltage is advisable.

(ii) Test Procedure

The d.c. source is normally generated from a mains or battery powered megaohmmeter of sufficient capacity, and the values recorded. The recommended level of voltage is 1 kv d.c. for windings up to and including 6.6kv and 5kv d.c. for winding of 11kv or greater.

(iii) Interpretation

If low values are obtained after correction to 40°C, serious moisture or other contamination is indicated, see 4.1.3 (iv) Further investigation is required followed by a dry-out, as necessary, prior to a.c. testing.

### **4.0 REVIEW**

The Executive Director, Operation Services, will be responsible for reviewing this document on a 2-yearly basis.

## **APPENDIX-I**

### **PROCEDURE FOR MEASUREMENT OF TAN DELTA WITH WATER IN GENERATOR WINDINGS**

Tan delta measurements on installed large generator stators, using a Schering bridge or other methods, are complicated by the presence of circulating cooling water in the stator winding. The losses in the low resistance water path to earth, in parallel with the insulation under test, are of the same order of magnitude as the dielectric losses and must be eliminated from measurements. Three methods of achieving this are theoretically possible.

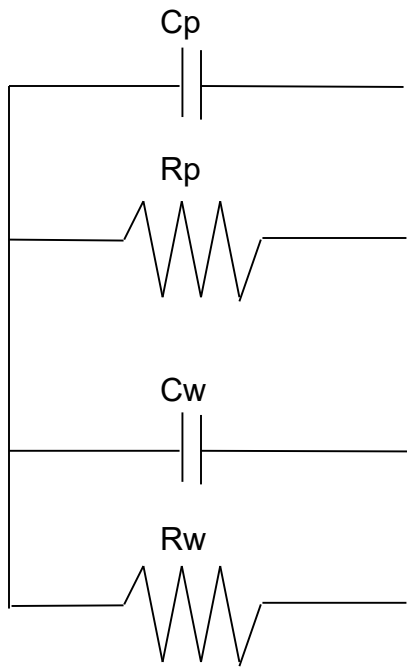
- First the water can be drained from the windings, but this can be a costly procedure as it is essential to remove all water droplets from the internal insulation surfaces.
- The losses in the water can be eliminated from the bridge measurements if suitable guard arrangements are used to bypass the leakage in the water from the measuring circuit. This may not be possible with the existing instrumentation.
- Losses in the water columns can be measured separately and deducted from the total measured losses.

The last method can be used successfully in large generators and it may be possible to predict tan delta from the "Wet" results to within an accuracy of  $\pm 3\%$  of the 'dry' or true values.

This new technique of measurement of tan delta has been successfully developed based on the fact that the a.c. and d.c resistance of deionised water column are almost identical.

## DESCRIPTION:

The equivalent circuit of the stator insulation and cooling water path are as shown below:



$$R_{TOT} = \frac{R_p \times R_w}{R_p + R_w}$$

$$\text{Or } \frac{1}{R_{TOT}} = \frac{1}{R_p} + \frac{1}{R_w}$$

$$C_{TOT} = C_p + C_w$$

$$\text{i.e Total } X_c = \frac{1}{2 \pi f (C_p + C_w)}$$

but as  $C_w \ll C_p$  then

$$\text{Total } X_c = \frac{1}{2 \pi f C_p}$$

Apparent insulation loss angle from a bridge measurement will be

$$= \frac{X_c}{R_{TOT}}$$

But true insulation loss angle

$$= \frac{X_c}{R_p}$$

This new technique thus depends on obtaining a correct value for  $R_p$  from the 'lumped'  $R_{TOT}$  inherent in the Schering Bridge measurement with circulating cooling water present. This is readily accomplished by the procedure outlined below:

1. From the schering bridge measurements evaluate the phase capacitance and apparent loss angle at each voltage.
2. From 1, calculate the equivalent resistance  $R_{TOT}$  of the water and insulation in parallel.
3. From a low voltage d.c. bridge measurement evaluate  $R_w$ . The paralleling d.c. resistance of the insulation of the order of Megohms and hence is eliminated from the measurement because it will appear as an open circuit to the d.c. bridge.
4. Evaluate  $R_p$  by parallel resistance considerations.
5. Use this value of  $R_p$  and the phase capacitance already evaluated to calculate the true loss angle at 50 Hz. Clearly the accuracy of this new technique depends on the correct prediction of  $R_p$  from measurements of  $R_w$  and  $R_{TOT}$ . An estimate of the accuracy of this prediction can be obtained using equation (1):

$$= \frac{1}{R_{TOT}} = \frac{1}{R_w} + \frac{1}{R_p} \quad \text{--- (1)}$$

$$\text{OR } \frac{1}{R_p} = \frac{1}{R_{TOT}} - \frac{1}{R_w}$$

Defferentiating:

$$\frac{\delta R_p}{R_p^2} = - \frac{d R_{TOT}}{R_{TOT}^2} + \frac{d R_w}{R_w^2} \quad \text{--- (2)}$$

$$\text{Let \% error in measuring } R_{TOT} \text{ be } E_{TOT}, \text{ i.e. } E_{TOT} = \frac{D R_{TOT}}{R_{TOT}} \times 100$$

Let % error in measuring  $R_w$  be  $E_w$ , i.e.  $E_w = \frac{\delta R_w}{R_w} \times 100$

Let % error in calculating  $R_p$  be  $E_p$ , i.e.  $E_p = \frac{\delta R_p}{R_p} \times 100$

Then from above 
$$\frac{E_p}{R_p} = \frac{E_{TOT}}{R_{TOT}} - \frac{E_w}{R_w}$$

or 
$$E_p = \frac{E_{TOT} R_p}{R_{TOT}} - \frac{E_w R_p}{R_w} \quad (3)$$

But from (1)

$$R_p = \frac{R_{TOT} R_w}{R_w - R_{TOT}}$$

Substituting in (3)

$$E_p = \frac{E_{TOT} R_w - E_w R_{TOT}}{R_w - R_{TOT}} \quad (4)$$

Depending upon the different designs of 500 MW Generator full of circulating stator cooling water has shown:

- a) Typically  $R_w$  is of the order of 1 M ohm.
- b) Typically  $R_{TOT}$  is of the order of 0.3 M ohm.
- b) Typically  $E_w$  is of the order of  $\pm 2\%$ .
- d) Typically  $E_{TOT}$  is of the order of  $\pm 2\%$ .

Substituting in (4)

$$E_p = \frac{\pm 2 \times 1 - \pm 2 \times 0.3}{0.7} - \pm 3\%$$

Therefore, by this new technique Rp can be predicated to approx.  $\pm 3\%$  accuracy.